

Ethanol Blending in Gasoline – Slovenia

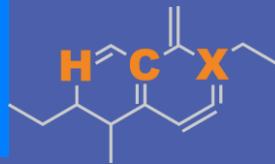
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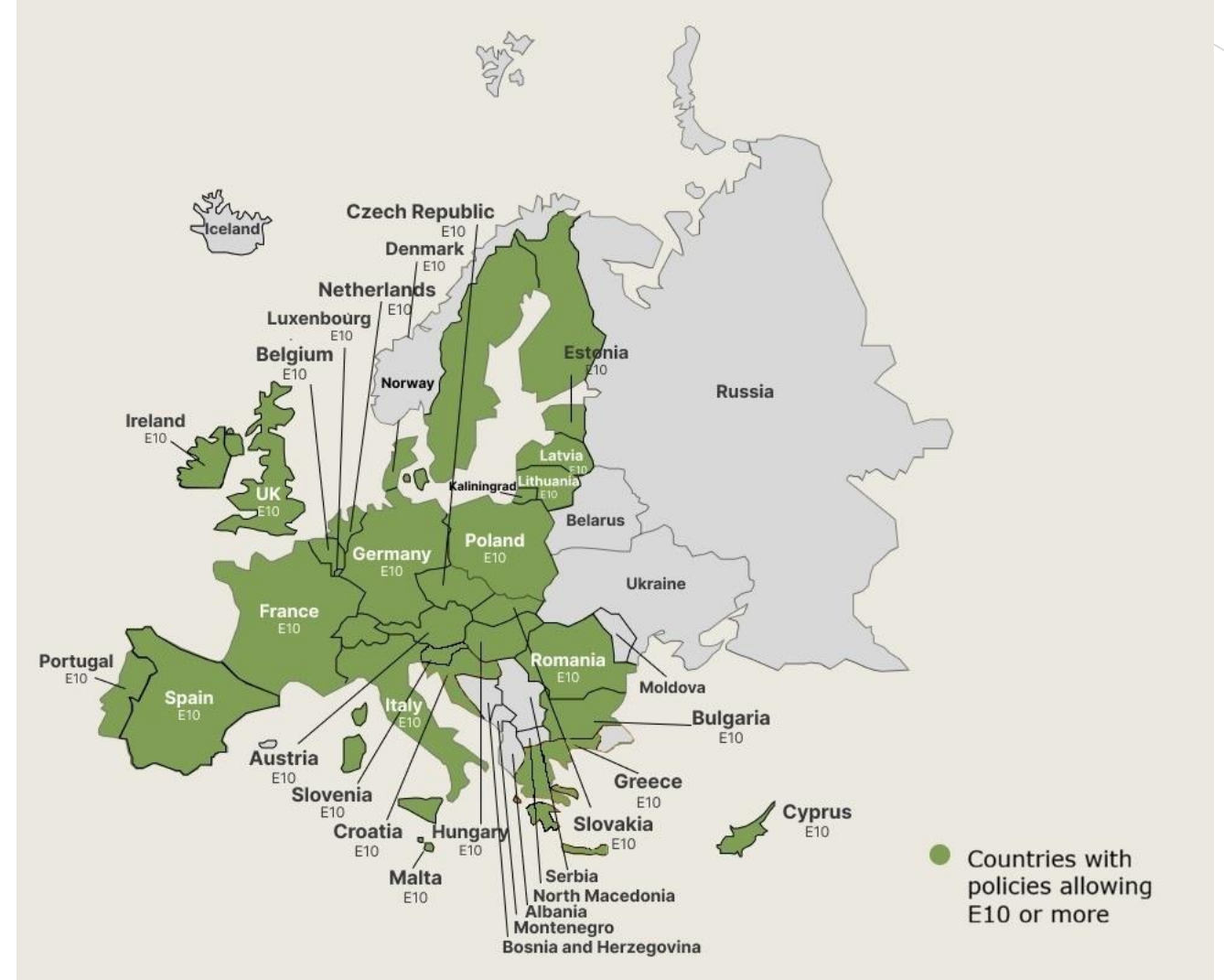


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Ethanol Blending in Europe

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

Ethanol Gasoline Blending in Slovenia

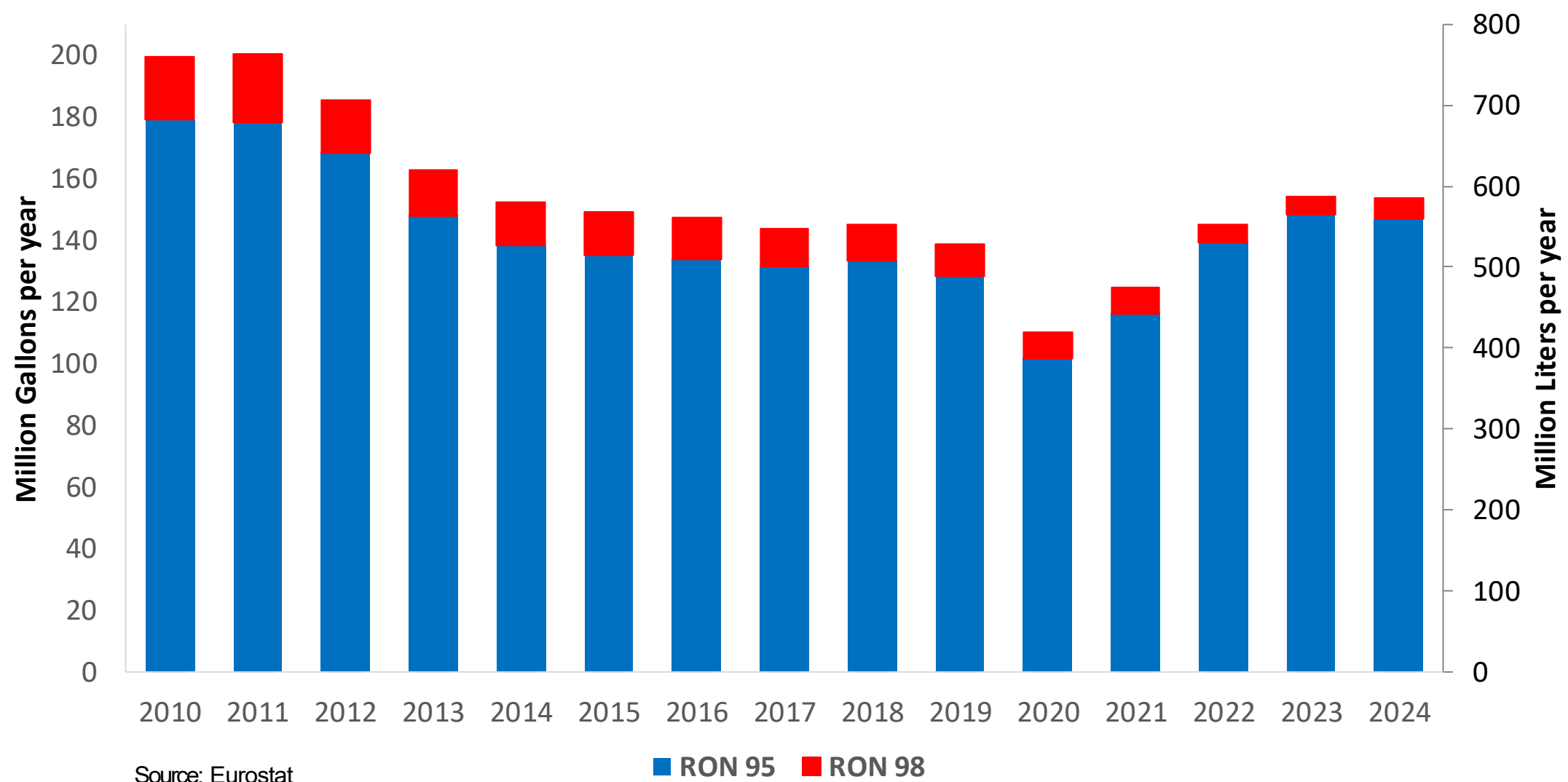


In 2024, gasoline consumption in Slovenia was 585 million liters (155 million gallons) and growth rate was 2.0%. Gasoline production and net imports were 0 and 574 million liters (0 and 152 million gallons) correspondingly. Slovenia complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.1 RON.

In 2024, ethanol consumption in Slovenia was 3 million liters (1 million gallons) representing 0.5% of gasoline demand. Ethanol production and Net Imports were 0 and 3 million liters (0 and 1 million gallons) correspondingly. Ethanol source is Cereals 100%.

Source: Eurostat, European Commission

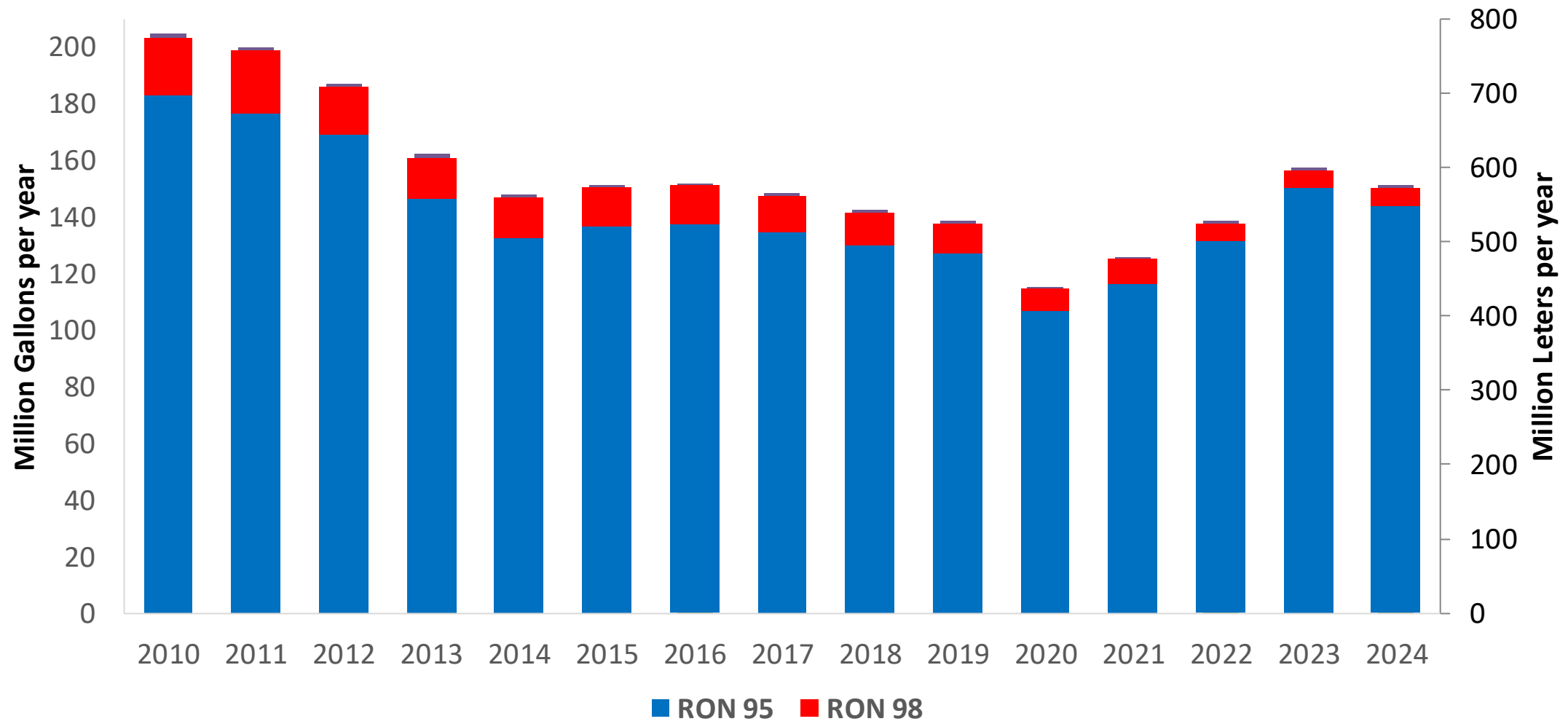
Gasoline Demand by Grade in Slovenia



Source: Eurostat

Gasoline Imports in Slovenia

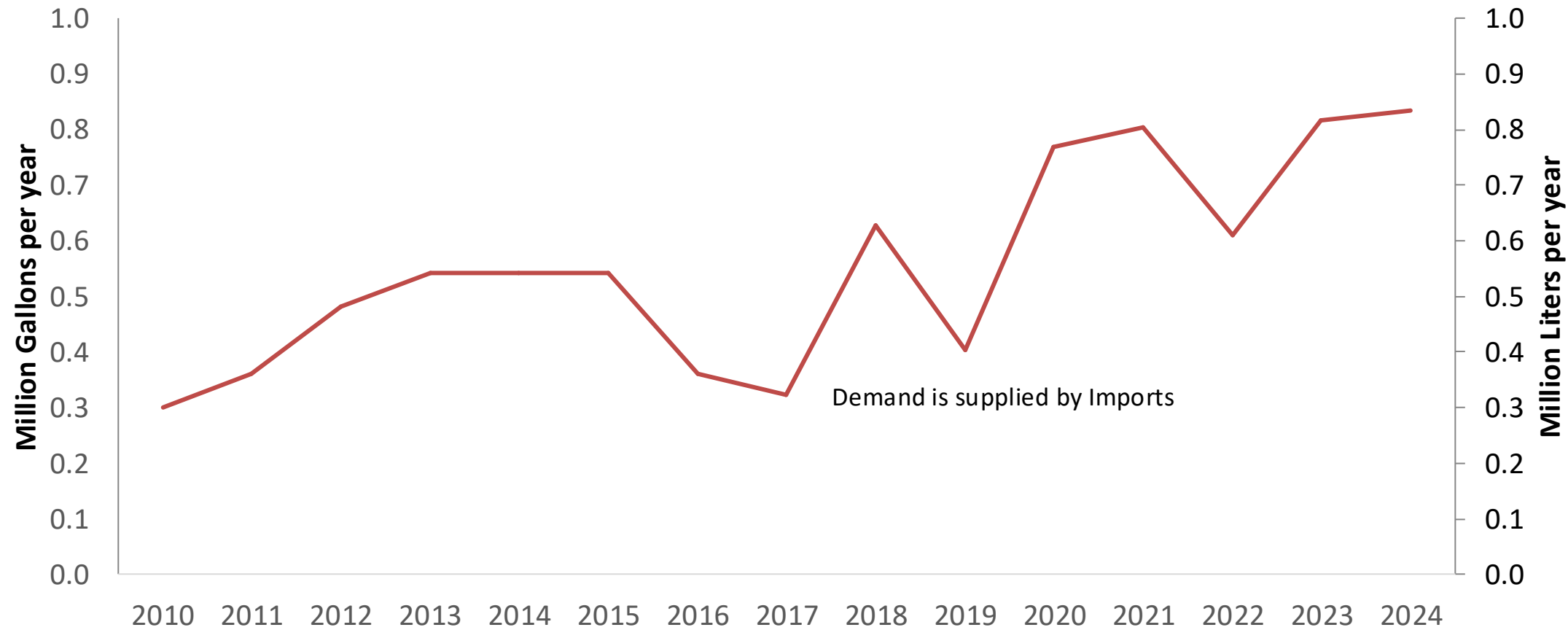
Gasoline Imports by Grade in Slovenia



Source: Eurostat

Ethanol Balance in Slovenia

Ethanol Demand in Slovenia



Source: Eurostat

Gasoline Quality in Slovenia

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Countries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	11.2							
Advanced Biofuels (%cal)	0.2							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commission

The FARO90 logo is shown in a blue box, followed by a chemical structure diagram of a substituted cyclohexene ring. The structure features a double bond between carbons 1 and 2, with substituents at various positions, including a methyl group at C3, a propyl group at C4, and a vinyl group at C5. The carbons are labeled with orange letters H, C, and X.

Catalytic Gasoline
Light Primary Naphta
Heavy Primary Naphta
Reformate
Isopentane
Alkylate
Isomerate
Hydrocracked Gasoline
Coker Naphta
n-Butane
MTBE
ETBE
TAME
Aromatics

The chart displays the percentage of respondents in various European countries who believe the EU is a good idea, broken down by age group. The countries are listed on the y-axis, and the percentage scale is on the x-axis from 0% to 100%.

The age groups represented by the segments are:

- 18-24 (Dark Blue)
- 25-34 (Red)
- 35-44 (Grey)
- 45-54 (Purple)
- 55-64 (Light Grey)
- 65-74 (Orange)
- 75-84 (Yellow)
- 85+ (Brown)
- 90+ (Dark Blue)

Key observations from the chart include:

- France:** Shows a significant portion of the 18-24 age group (approx. 20%) and a smaller portion of the 90+ age group (approx. 5%).
- Germany:** Shows a significant portion of the 18-24 age group (approx. 15%) and a smaller portion of the 90+ age group (approx. 5%).
- Italy:** Shows a significant portion of the 18-24 age group (approx. 15%) and a smaller portion of the 90+ age group (approx. 5%).
- Spain:** Shows a significant portion of the 18-24 age group (approx. 15%) and a smaller portion of the 90+ age group (approx. 5%).
- UK:** Shows a significant portion of the 18-24 age group (approx. 10%) and a smaller portion of the 90+ age group (approx. 5%).

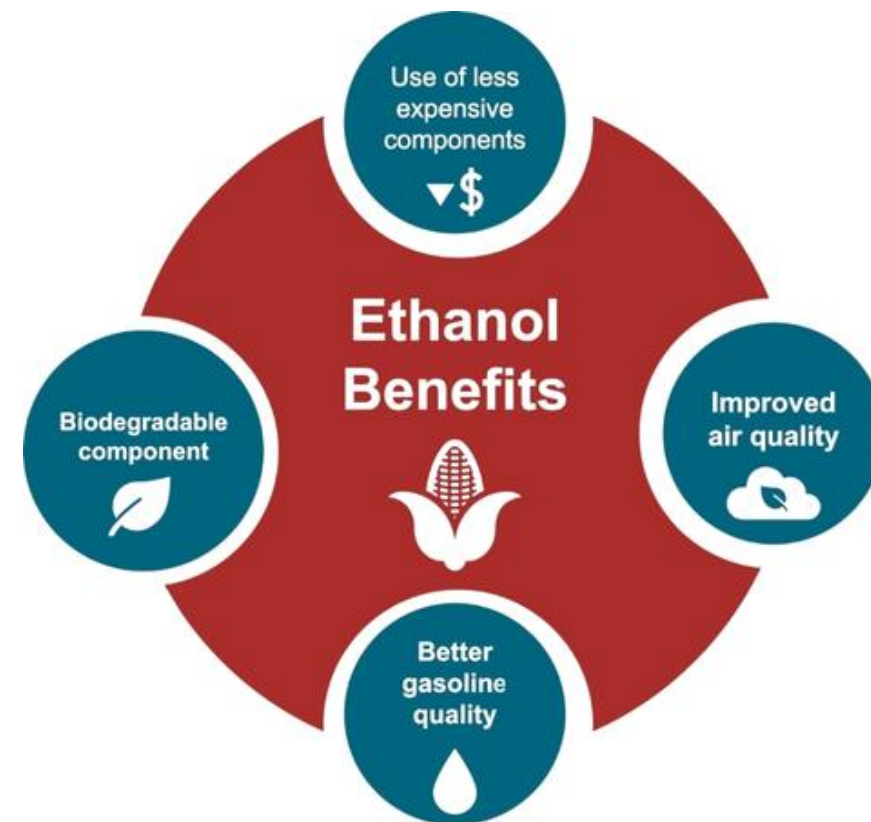
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

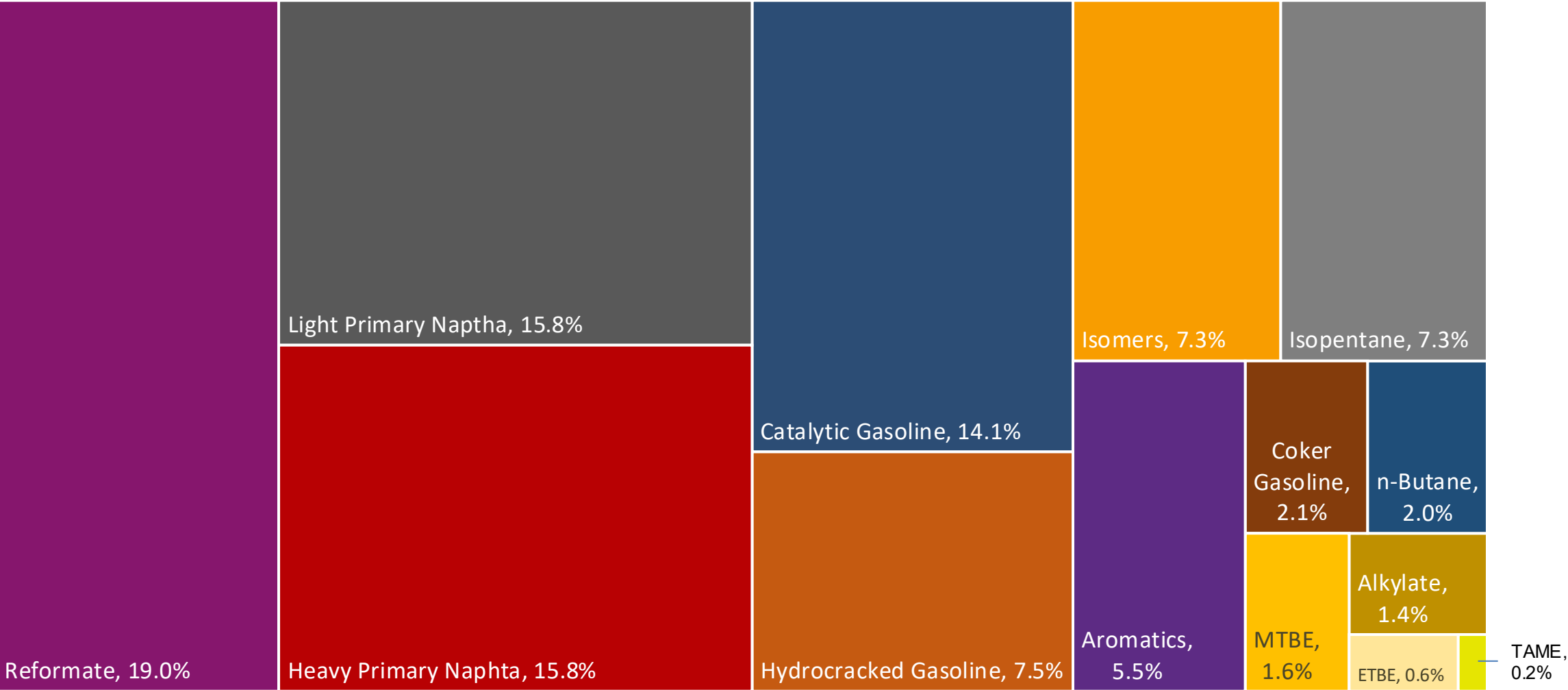
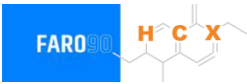
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

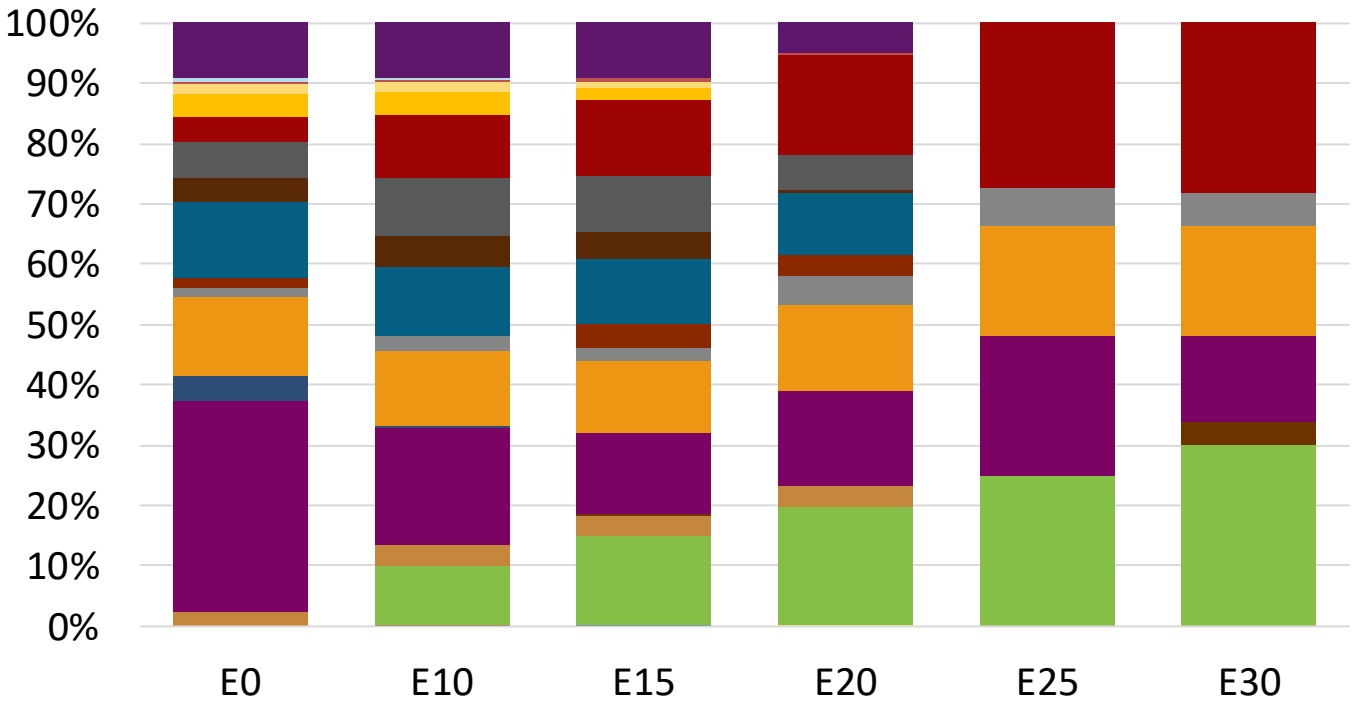
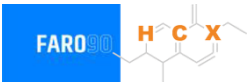
The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Slovenia Available Blending Components



Slovenia – Gasoline 95 – Constant Octane

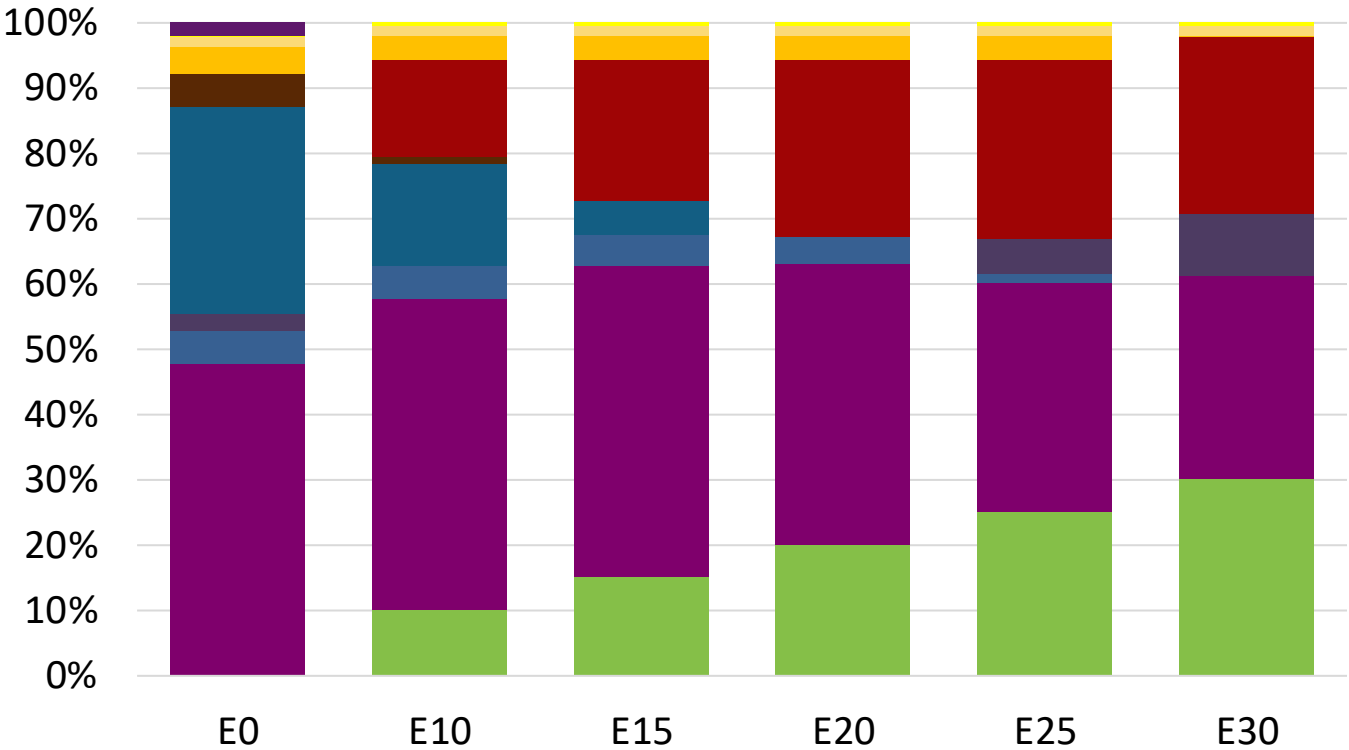


Average CIF 2024
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.1	95.6
Price (US\$/gal)	\$ 2.292	\$ 2.164	\$ 2.103	\$ 2.022	\$ 1.900	\$ 1.834

Source: Faro90

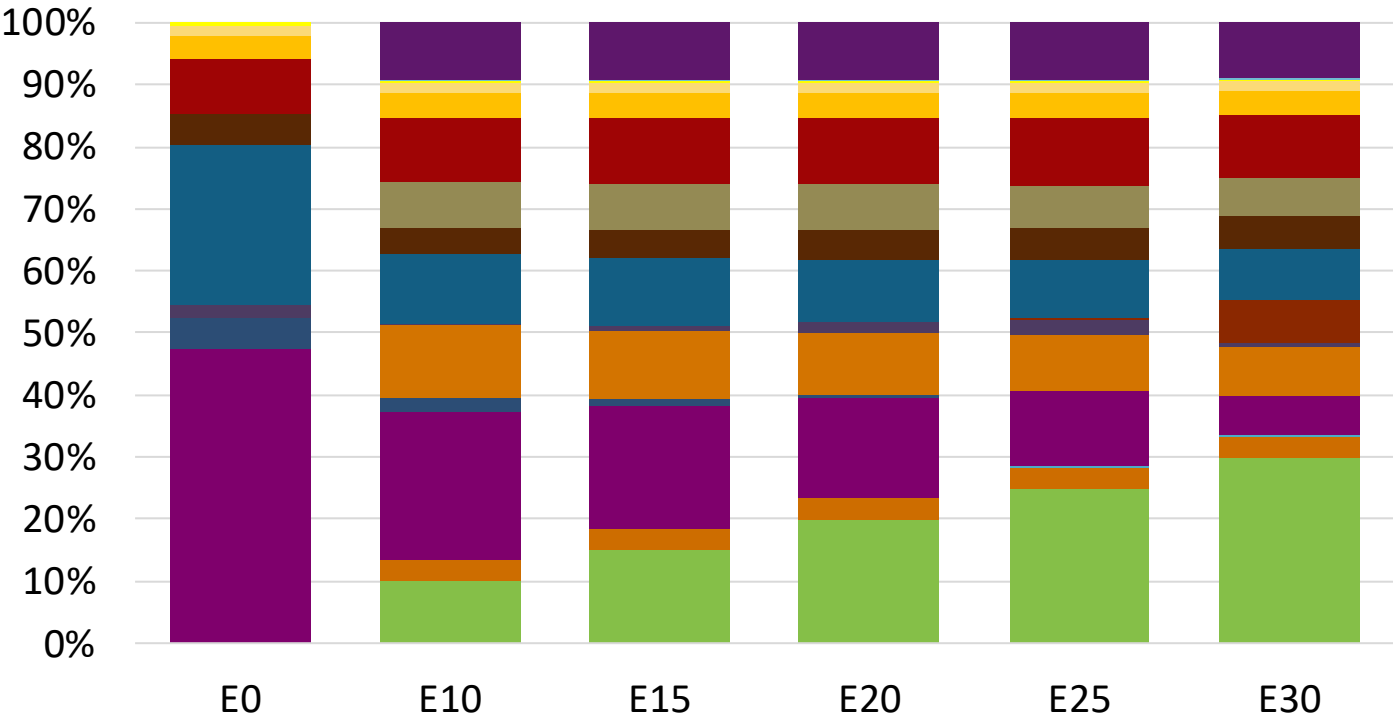
Slovenia - Gasoline 98 - Constant Octane



Average CIF 2024 Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.367	\$ 2.238	\$ 2.166	\$ 2.089	\$ 2.012	\$ 1.945

Slovenia – Gasoline 95 – Increased Octane

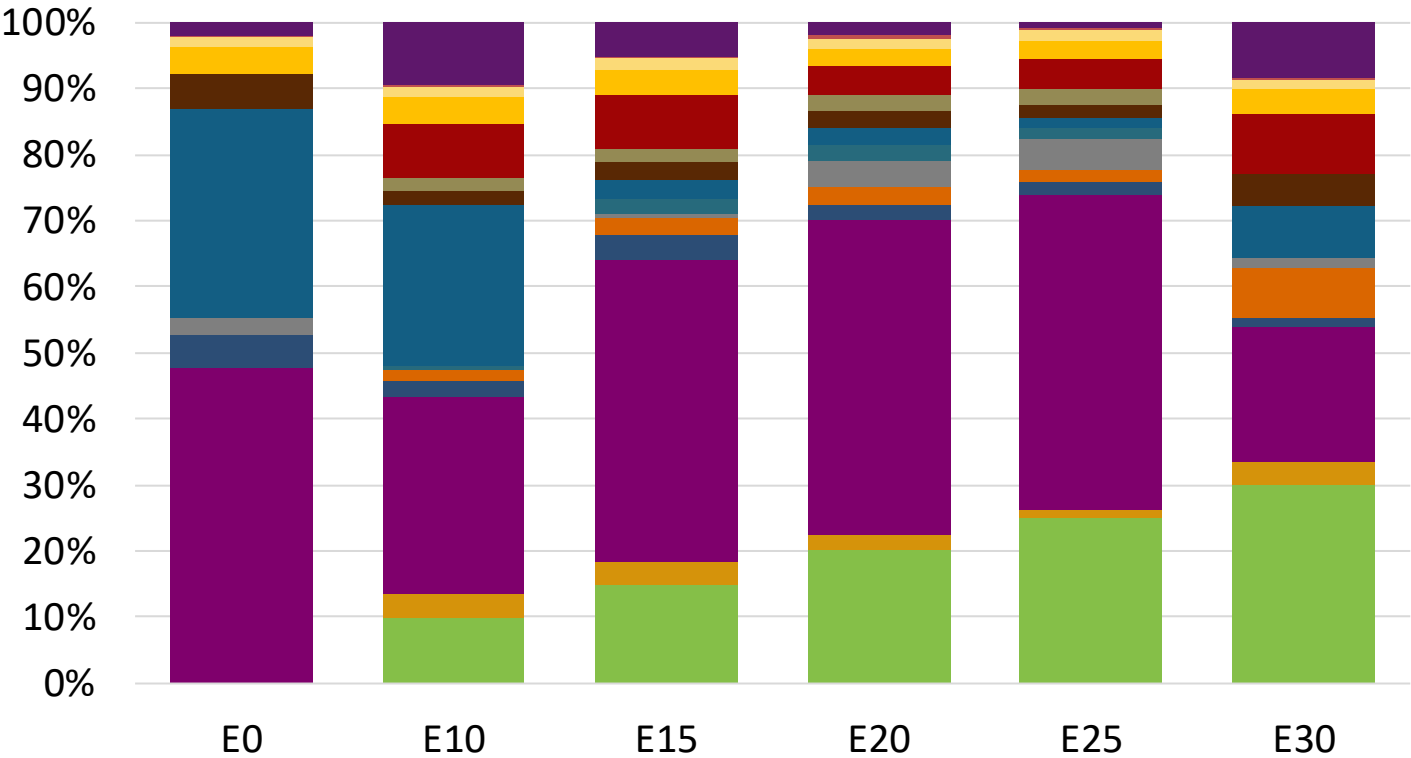
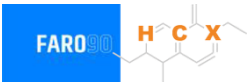


Average CIF 2024
Prices

Octane (RON)	95.0	96.5	97.9	99.4	100.8	102.0
Price (US\$/gal)	\$ 2.229	\$ 2.229	\$ 2.229	\$ 2.229	\$ 2.229	\$ 2.229

Source: Faro90

Slovenia – Gasoline 98 – Increased Octane



Average CIF 2024
Prices

Octane (RON)	98.0	99.2	101.3	102.6	104.0	104.7
Price (US\$/gal)	\$ 2.367	\$ 2.367	\$ 2.367	\$ 2.367	\$ 2.367	\$ 2.367

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

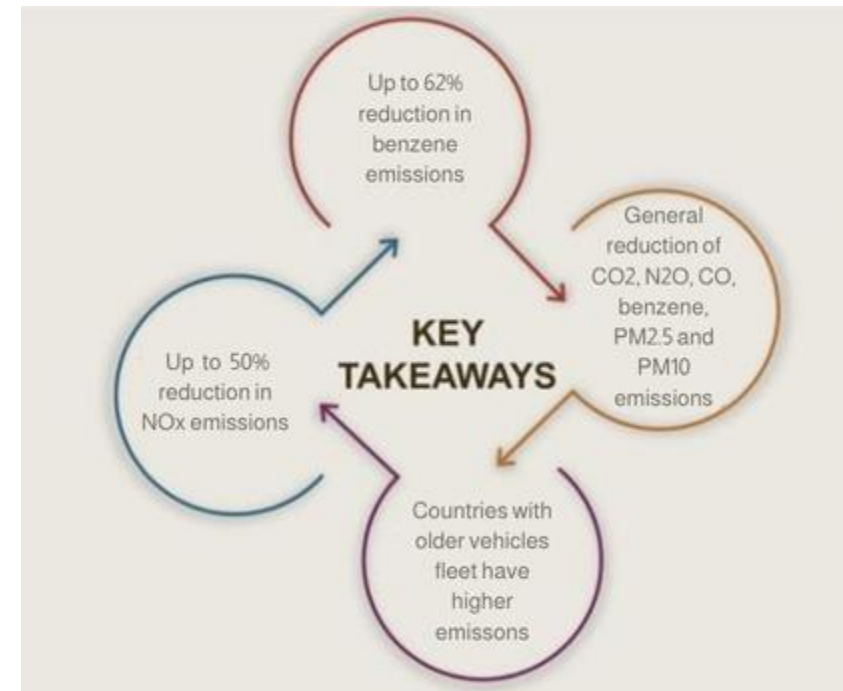
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

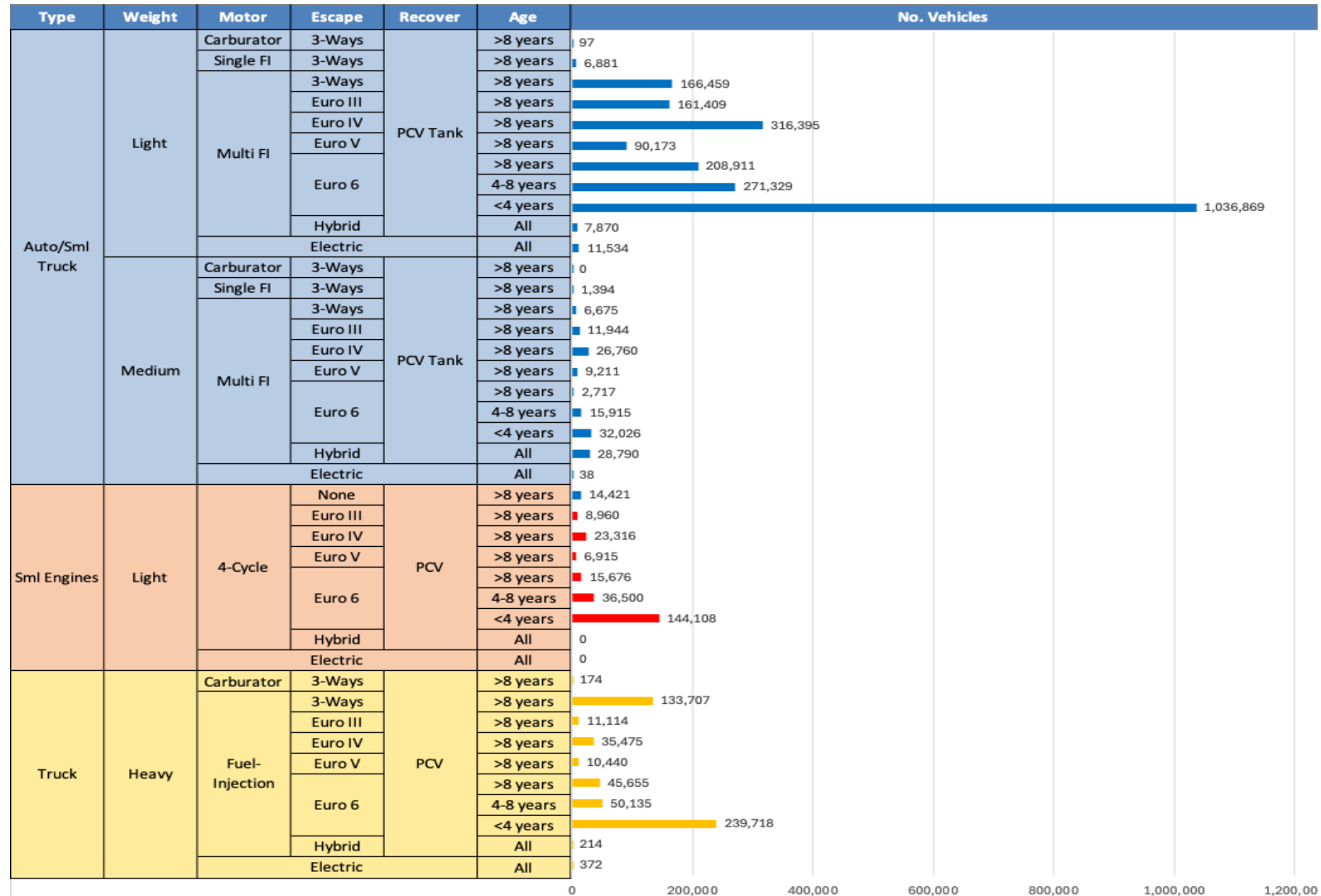
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

**Source: Bureau of transportation statistics.*

Main Results



Gasoline Vehicle Fleet - Slovenia

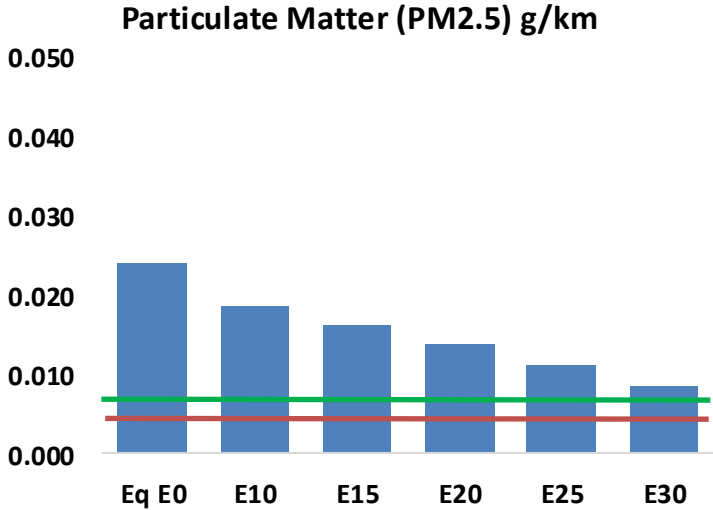
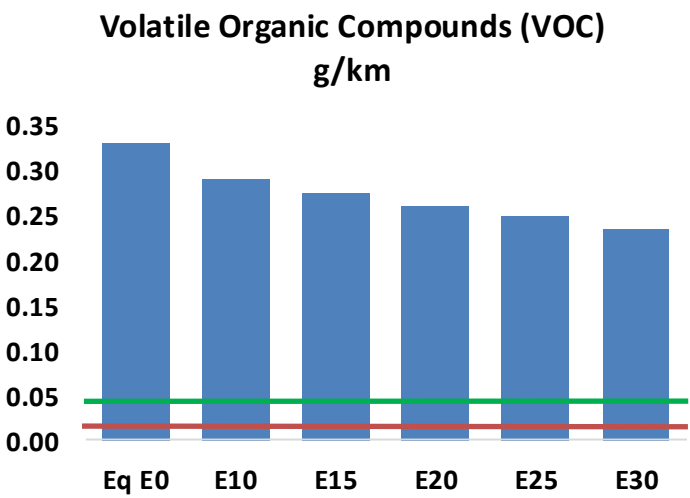
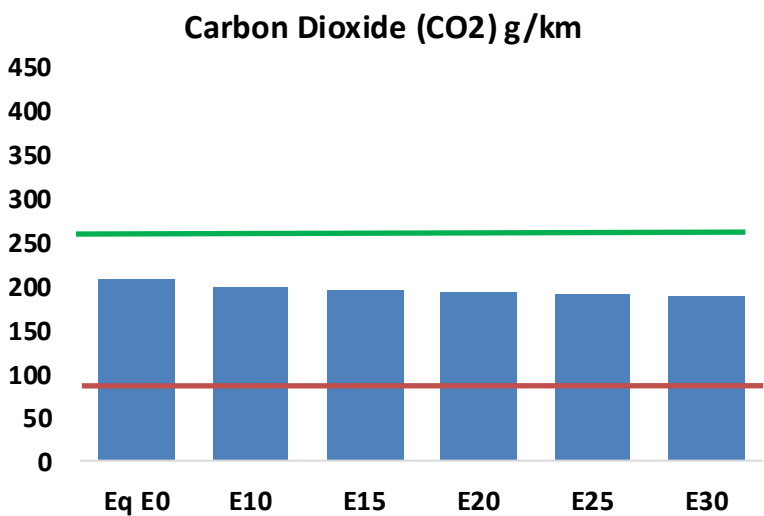
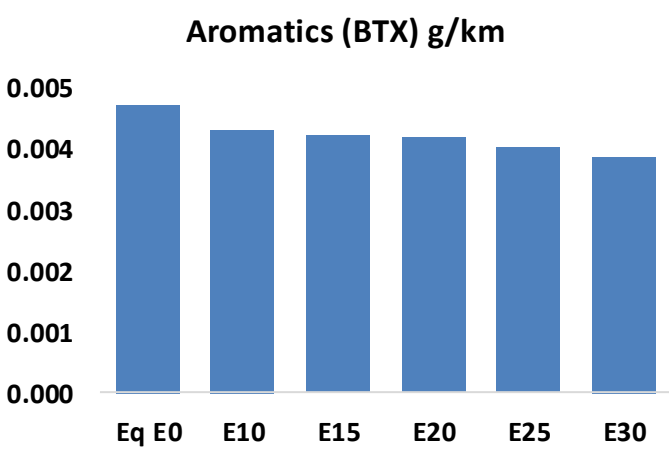
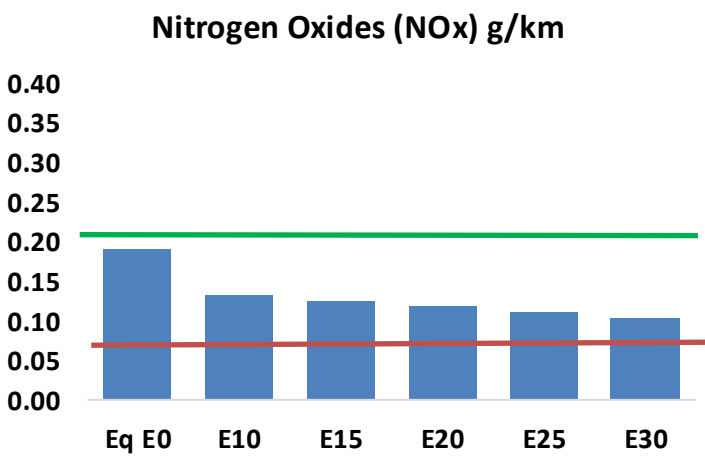
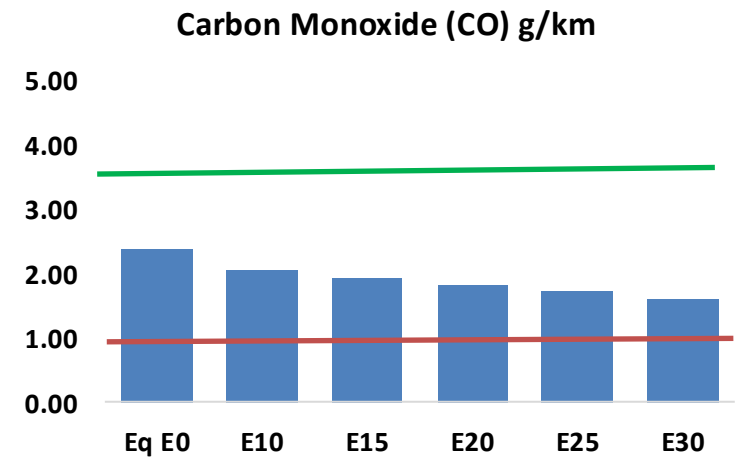
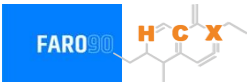


Vehicle Fleet: 3,190,297
Gasoline Vehicle Fleet: 1,163,863

Source: Eurostat, ACEA

Slovenia – Vehicular Emissions

TIER III BIN 125 United States
Euro 6 Standard



Slovenia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	66,460	62,046	57,631	54,148	50,707	48,079	45,138	-13%	-24%
CO2	5,814,866	5,665,340	5,524,122	5,413,058	5,358,100	5,329,401	5,231,165	-5%	-8%
VOC	9,254	8,696	8,137	7,709	7,293	6,976	6,539	-12%	-21%
NOx	5,327	3,996	3,729	3,515	3,315	3,094	2,861	-30%	-38%
SOx	65	60	55	51	47	43	38	-15%	-28%
NH3	2,043	2,023	2,003	2,012	2,035	2,051	2,064	-2%	0%
N2O	61	61	60	61	62	63	63	-1%	2%
CH4	2,030	2,030	2,030	2,061	2,105	2,144	2,180	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	63	61	59	57	57	56	55	-6%	-9%
Acetaldehyde	160	183	269	410	558	638	754	68%	249%
Formaldehyde	545	529	617	725	759	840	914	13%	39%
Benzene	132	127	120	119	118	112	109	-9%	-11%
PM2.5	371	329	288	250	212	172	130	-22%	-43%
PM10	303	269	235	204	173	141	107	-22%	-43%

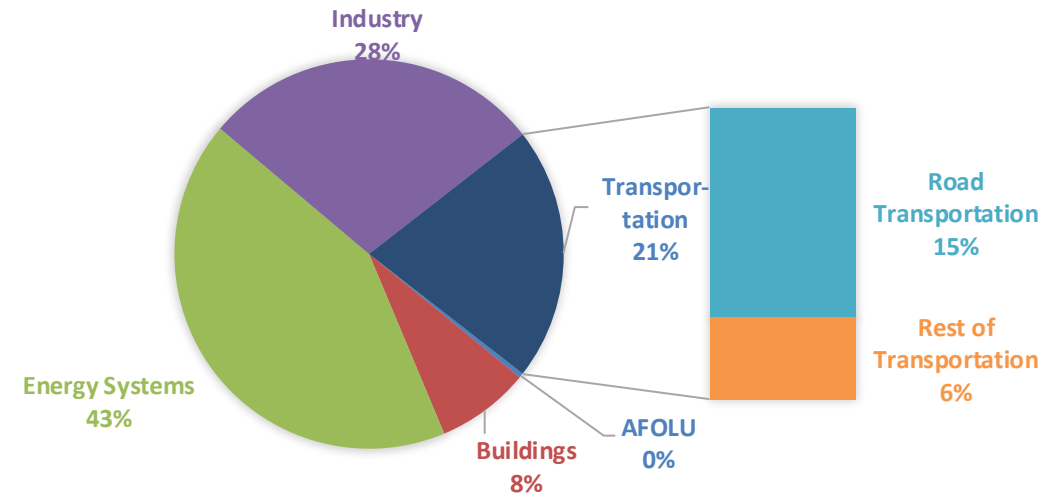
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

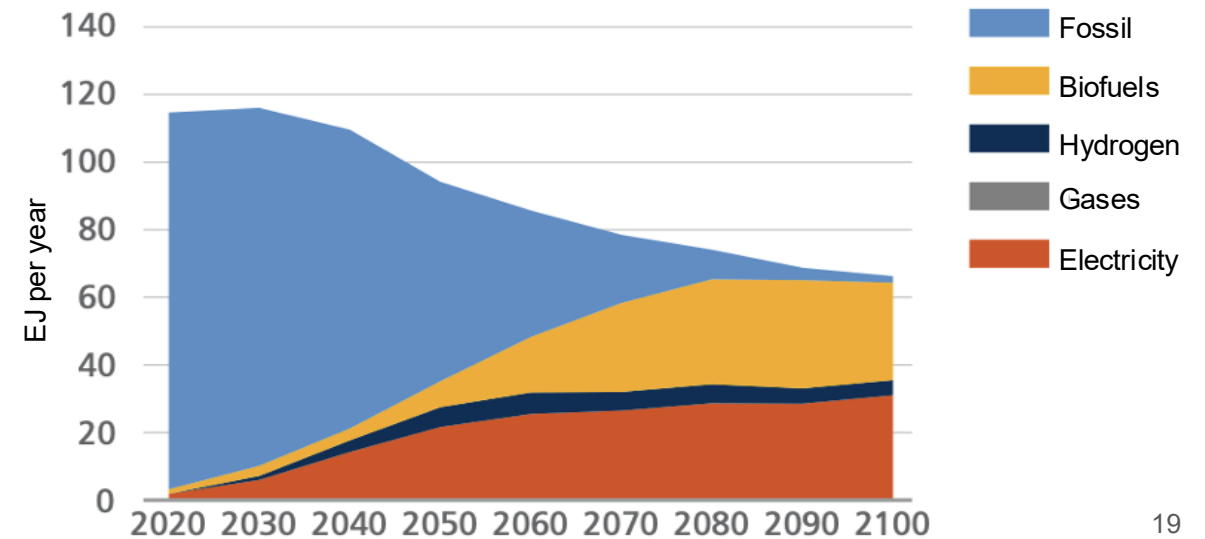
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

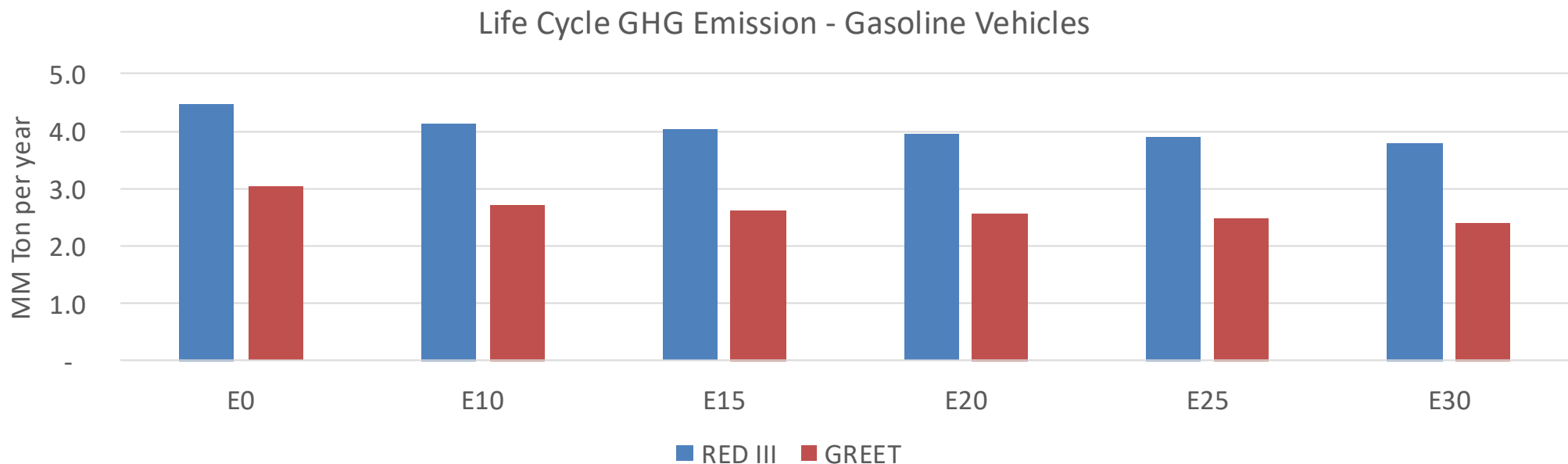
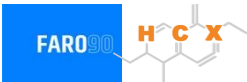
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Slovenia – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	16.5
Target @ 2035 (MMT/year)	9.3
Delta	7.2

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	10.8%	13.7%	16.0%	18.0%	20.9%
RED II/III	0.0%	7.7%	9.9%	11.6%	13.1%	15.2%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	4.6%	5.8%	6.7%	7.6%	8.9%
RED II/III	0.0%	4.8%	6.2%	7.2%	8.1%	9.4%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90